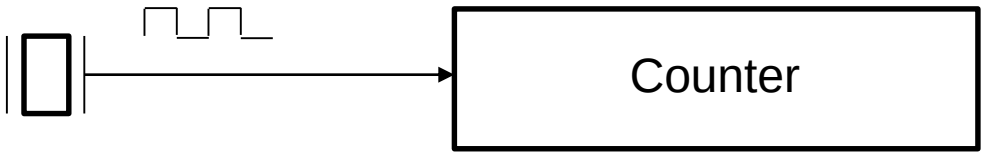
Computer interfacing (CI)

6. Timers

6.1 Need of timers

Time is represented by the combination of a **clock source** and a digital **binary counter**.

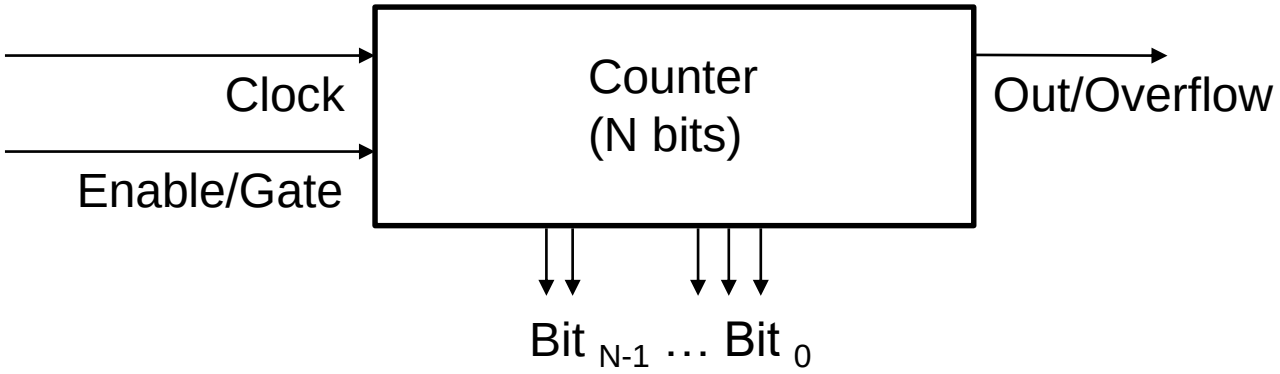


There are several applications that cannot be implemented without a timer:

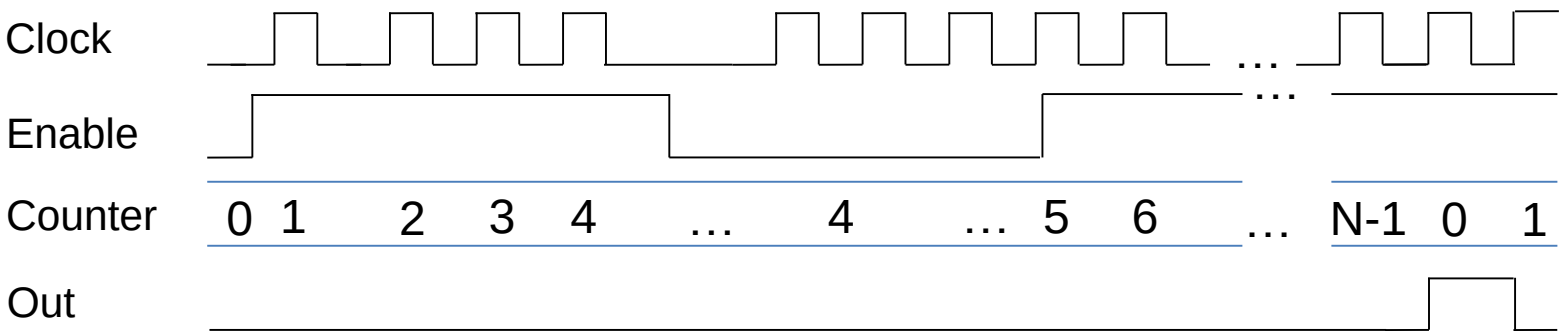
- Periodic interrupt generation (system clock, user clock, calendars…)
- Event arrival time recording and comparison. Time references, timestamps.
- Generation of waveforms with certain frequency and duty cycle.
- Pulse width and period measurement.
- Frequency and duty cycle measurement

6.1 Need of timers

Generic timer structure:



Functional concepts:



6.2 PIC18 Timer system.

A PIC18F45K22 microcontroller has up to 7 timers: Timer 0…Timer 6.

- Timer 0, Timer 1, Timer 3 and Timer 5 are 16-bit timers.
- Timer 2, Timer 4 and Timer 6 are 8-bit.

- When a timer rolls over (overflow), an interrupt may be generated if it is enabled.

- Timer 2, Timer 4 and Timer 6 use instruction cycle clock as the clock source whereas the other timers may also use external clock input as the clock source.

- All the timers have an enable mechanism.

- Timer 0 is designed to act as a time base (core interrupt) while the other timers are in the peripheral group (alternate time base and CCP operation).

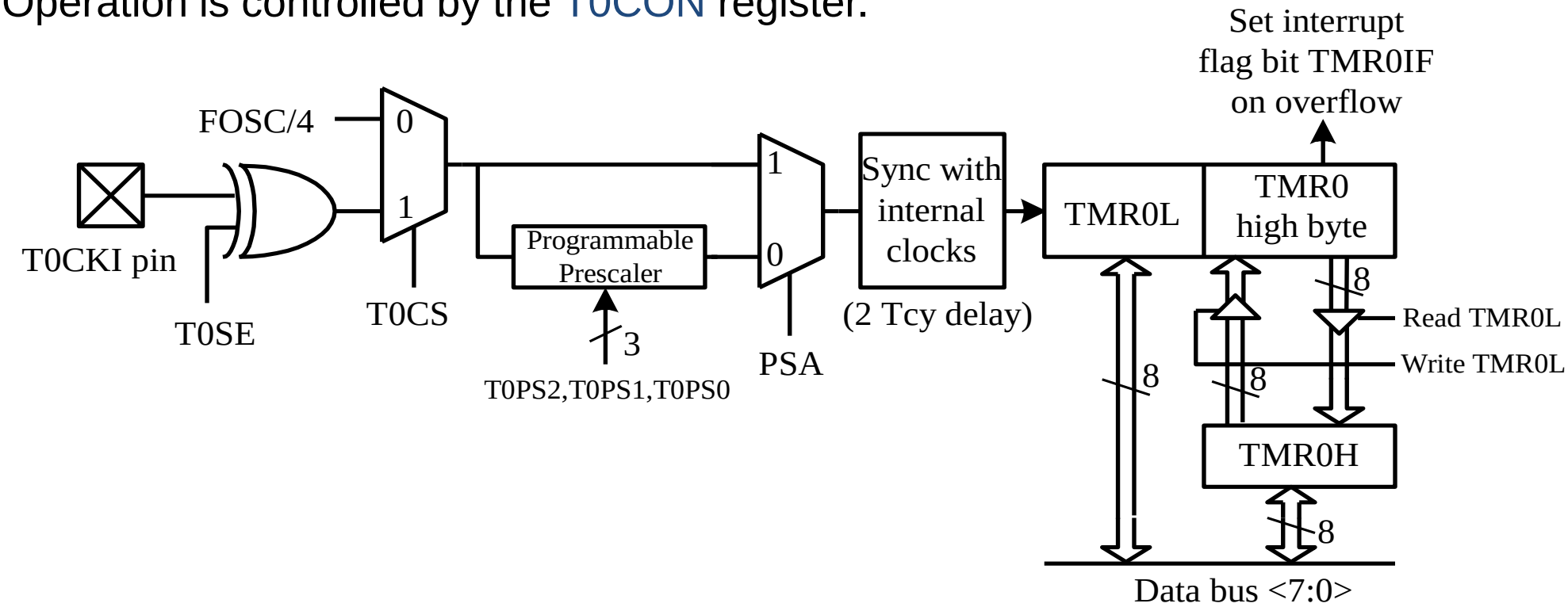
6.3 Timer 0

Can be configured as 8-bit or 16-bit. 16 bit mode, requires bus arbitration (H/L).

Is a timer/counter depending upon the clock source.

An interrupt (TMR0IF) may be requested when Timer0 rolls over from 0xFFFF to 0x0000.

Operation is controlled by the T0CON register.



6.3 Timer 0

T0CON register, bit description.

Bit value at reset (example=1)

Operation (read/write)

Bit / bit group name

R/W-1	R/W-1	R/W-1	R/W-1	R/W-1	R/W-1	R/W-1	R/W-1
TMR0ON	T08BIT	T0CS	T0SE	PSA		T0PS<2:0>	
bit 7							bit 0

bit 7 **TMR0ON:** Timer0 On/Off Control bit
1 = Enables Timer0
0 = Stops Timer0

bit 6 **T08BIT**: Timer0 8-bit/16-bit Control bit
1 = Timer0 is configured as an 8-bit timer/counter
0 = Timer0 is configured as a 16-bit timer/counter

bit 5 **T0CS**: Timer0 Clock Source Select bit
1 = Transition on T0CKI pin
0 = Internal instruction cycle clock (CLKOUT)

bit 4 **T0SE**: Timer0 Source Edge Select bit
1 = Increment on high-to-low transition on T0CKI pin
0 = Increment on low-to-high transition on T0CKI pin

bit 3 **PSA**: Timer0 Prescaler Assignment bit
1 = TImer0 prescaler is NOT assigned. Timer0 clock input bypasses prescaler.
0 = Timer0 prescaler is assigned. Timer0 clock input comes from prescaler output.

bit 2-0 **T0PS<2:0>**: Timer0 Prescaler Select bits
111 = 1:256 prescale value
110 = 1:128 prescale value
101 = 1:64 prescale value
100 = 1:32 prescale value
011 = 1:16 prescale value
010 = 1:8 prescale value
001 = 1:4 prescale value
000 = 1:2 prescale value

6.4 Timers 1, 3 and 5

These devices are 16-bit timer/counter depending upon the clock source.

An interrupt (TMRxIF) may be requested when Timerx rolls over from 0xFFFF to 0x0000.

These timers have a number of frequency input sources and a complex gate enable circuitry (if TMRxGE=1)

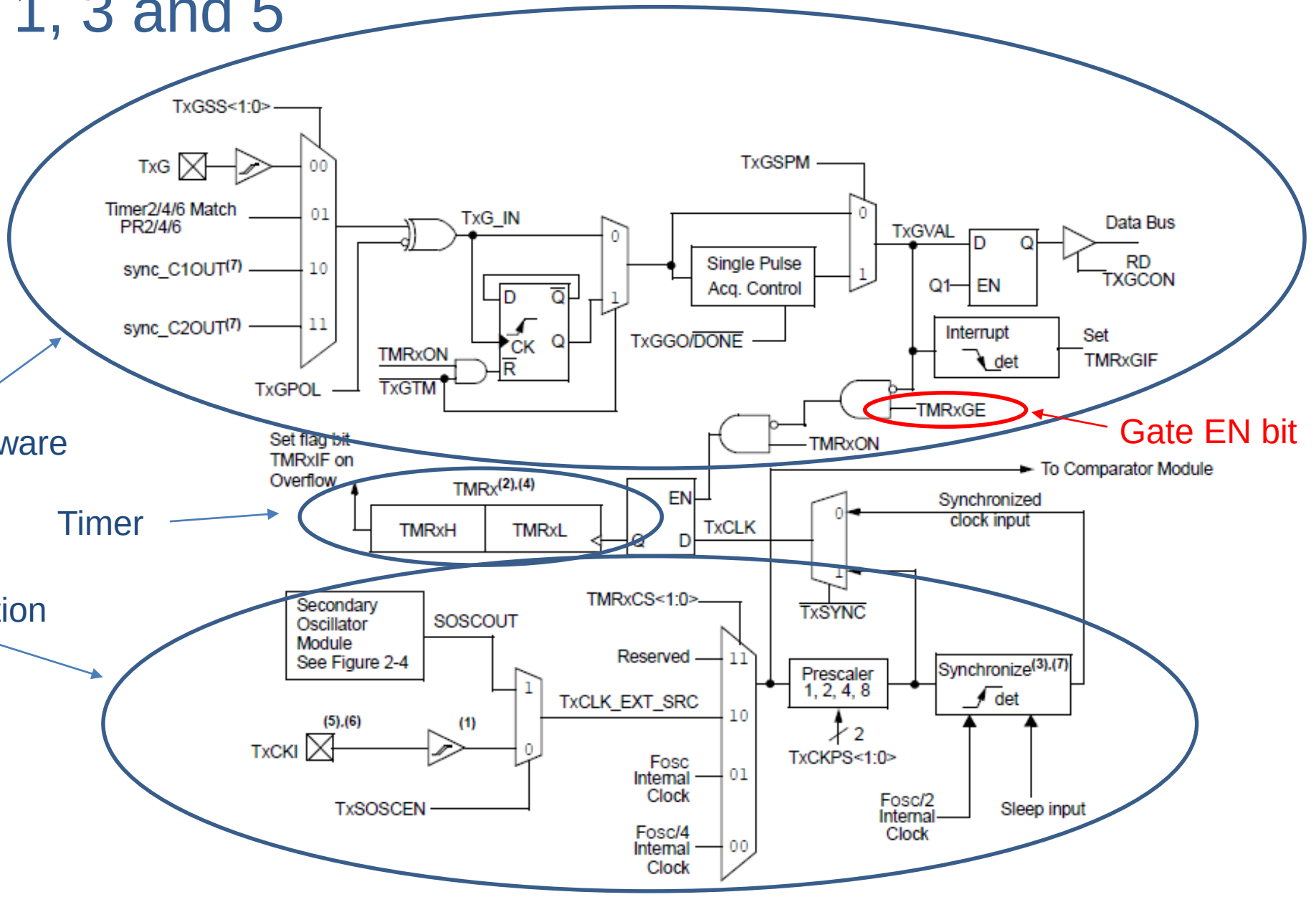
Timerx operation is controlled by two registers: the TxCON and TxGCON.

x={1,3,5}

These timers can be used to create time delays and measure the frequency of an unknown signal (being used by the CCP modules).

6.4 Timers 1, 3 and 5

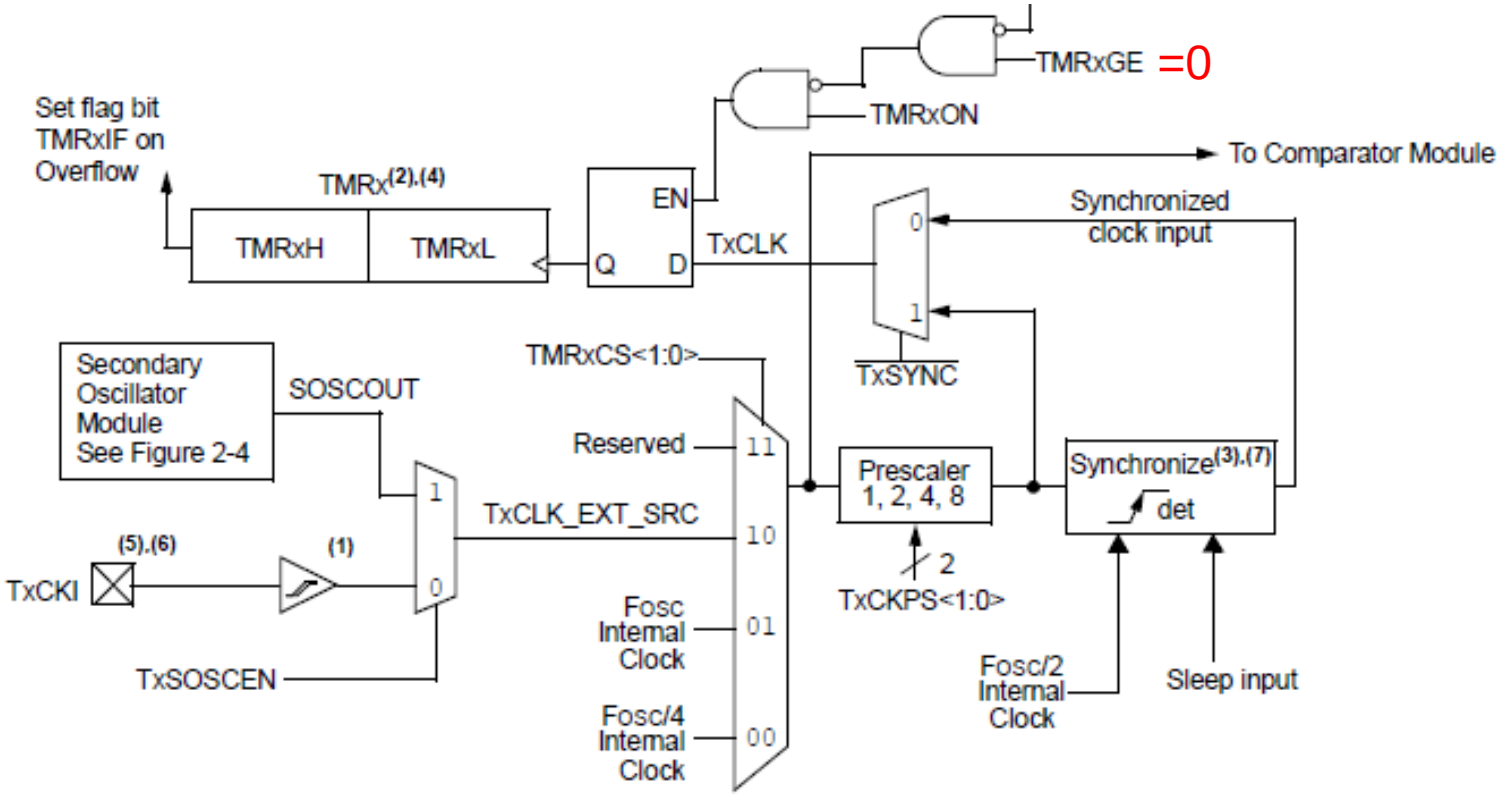
Block diagram:



6.4 Timers 1, 3 and 5

With TMRxGE=0 we have a standard Timer device. Gate hardware is ignored.

- TxCON register (x={1,3,5}) manages Timerx configuration.
- TxGCON register manages Gate Hardware configuration.



6.4 Timers 1, 3 and 5

TxCON Register:

x={1,3,5}

R/W-0/u	R/W-0/u	R/W-0/u	R/W-0/u	R/W-0/u	R/W-0/u	R/W-0/0	R/W-0/u
TMRxCS<1:0>		TxCKPS<1:0>		TxSOSCEN	$\overline{\text{TxSYNC}}$	TxRD16	TMRxON
bit 7							bit 0

bit 7-6 **TMRxCS<1:0>:** Timer1/3/5 Clock Source Select bits
11 = Reserved. Do not use.
10 = Timer1/3/5 clock source is pin or oscillator:
If TxSOSCEN = 0:
External clock from TxCKI pin (on the rising edge)
If TxSOSCEN = 1:
Crystal oscillator on SOSCI/SOSCO pins
01 = Timer1/3/5 clock source is system clock (FOSC)
00 = Timer1/3/5 clock source is instruction clock (FOSC/4)

bit 5-4 **TxCKPS<1:0>:** Timer1/3/5 Input Clock Prescale Select bits
11 = 1:8 Prescale value
10 = 1:4 Prescale value
01 = 1:2 Prescale value
00 = 1:1 Prescale value

bit 3 **TxSOSCEN:** Secondary Oscillator Enable Control bit
1 = Dedicated Secondary oscillator circuit enabled
0 = Dedicated Secondary oscillator circuit disabled

bit 2 **$\overline{\text{TxSYNC}}$:** Timer1/3/5 External Clock Input Synchronization Control bit
TMRxCS<1:0> = 1X
1 = Do not synchronize external clock input
0 = Synchronize external clock input with system clock (FOSC)

TMRxCS<1:0> = 0X
This bit is ignored. Timer1/3/5 uses the internal clock when TMRxCS<1:0> = 1X.

bit 1 **TxRD16:** 16-Bit Read/Write Mode Enable bit
1 = Enables register read/write of Timer1/3/5 in one 16-bit operation
0 = Enables register read/write of Timer1/3/5 in two 8-bit operation

bit 0 **TMRxON:** Timer1/3/5 On bit
1 = Enables Timer1/3/5
0 = Stops Timer1/3/5
Clears Timer1/3/5 Gate flip-flop

6.4 Timers 1, 3 and 5

TxGCON Register:

x={1,3,5}

R/W-0/u	R/W-0/u	R/W-0/u	R/W-0/u	R/W/HC-0/u	R-x/x	R/W-0/u	R/W-0/u
TMRxGE	TxGPOL	TxGTM	TxGSPM	TxGGO/$\overline{\text{DONE}}$	TxGVAL	TxGSS<1:0>	
bit 7							bit 0

bit 7 **TMRxGE:** Timer1/3/5 Gate Enable bit
If TMRxON = 0:
This bit is ignored
If TMRxON = 1:
1 = Timer1/3/5 counting is controlled by the Timer1/3/5 gate function
0 = Timer1/3/5 counts regardless of Timer1/3/5 gate function

bit 6 **TxGPOL:** Timer1/3/5 Gate Polarity bit
1 = Timer1/3/5 gate is active-high (Timer1/3/5 counts when gate is high)
0 = Timer1/3/5 gate is active-low (Timer1/3/5 counts when gate is low)

bit 5 **TxGTM:** Timer1/3/5 Gate Toggle Mode bit
1 = Timer1/3/5 Gate Toggle mode is enabled
0 = Timer1/3/5 Gate Toggle mode is disabled and toggle flip-flop is cleared
Timer1/3/5 gate flip-flop toggles on every rising edge.

bit 4 **TxGSPM:** Timer1/3/5 Gate Single-Pulse Mode bit
1 = Timer1/3/5 gate Single-Pulse mode is enabled and is controlling Timer1/3/5 gate
0 = Timer1/3/5 gate Single-Pulse mode is disabled

bit 3 **TxGGO/$\overline{\text{DONE}}$:** Timer1/3/5 Gate Single-Pulse Acquisition Status bit
1 = Timer1/3/5 gate single-pulse acquisition is ready, waiting for an edge
0 = Timer1/3/5 gate single-pulse acquisition has completed or has not been started
This bit is automatically cleared when TxGSPM is cleared.

bit 2 **TxGVAL:** Timer1/3/5 Gate Current State bit
Indicates the current state of the Timer1/3/5 gate that could be provided to TMRxH:TMRxL.
Unaffected by Timer1/3/5 Gate Enable (TMRxGE).

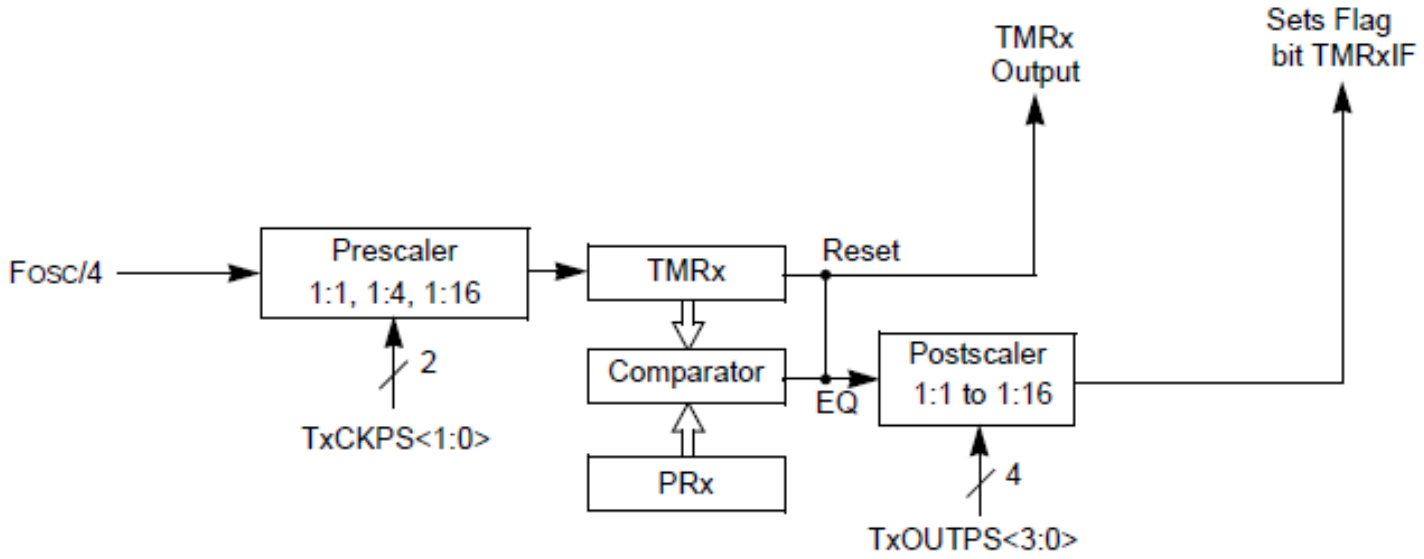
bit 1-0 **TxGSS<1:0>:** Timer1/3/5 Gate Source Select bits
00 = Timer1/3/5 Gate pin
01 = Timer2/4/6 Match PR2/4/6 output (See Table 12-5 for proper timer match selection)
10 = Comparator 1 optionally synchronized output (sync_C1OUT)
11 = Comparator 2 optionally synchronized output (sync_C2OUT)

6.5 Timers 2, 4 and 6

There are three 8-bit timers with instruction clock source Fcy (Fosc/4) and prescaler and postscaler block logic.

Controlled by TxCON and related to PRx registers (x={2,4,6}).
An interrupt may be requested when TMRx value matches PRx.

They can be a source for PWM signals (combined with CCP modules).



6.5 Timers 2, 4 and 6

TxCON Register:

x={2,4,6}

U-0	R/W-0	R/W-0	R/W-0	R/W-0	R/W-0	R/W-0	R/W-0
—	TxOUTPS<3:0>				TMRxON	TxCKPS<1:0>	
bit 7							bit 0

bit 7 **Unimplemented:** Read as '0'

bit 6-3 **TxOUTPS<3:0>:** TimerX Output Postscaler Select bits

0000 = 1:1 Postscaler
0001 = 1:2 Postscaler
0010 = 1:3 Postscaler
0011 = 1:4 Postscaler
0100 = 1:5 Postscaler
0101 = 1:6 Postscaler
0110 = 1:7 Postscaler
0111 = 1:8 Postscaler
1000 = 1:9 Postscaler
1001 = 1:10 Postscaler
1010 = 1:11 Postscaler
1011 = 1:12 Postscaler
1100 = 1:13 Postscaler
1101 = 1:14 Postscaler
1110 = 1:15 Postscaler
1111 = 1:16 Postscaler

bit 2 **TMRxON:** TimerX On bit

1 = TimerX is on
0 = TimerX is off

bit 1-0 **TxCKPS<1:0>:** Timer2-type Clock Prescale Select bits

00 = Prescaler is 1
01 = Prescaler is 4
1x = Prescaler is 16